Nishad Mansoor

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Education

Northeastern University — Boston, MA
MS in Data Science — *Expected December 2026*

Rutgers University — New Brunswick, NJ
BS in Finance and Business Analytics/Information Technology — *May 2024*

Experience

AI Engineer Intern — June 2026 – September 2026
EisnerAmper — *Iselin, NJ*

- Built a pipeline tracking AI usage across the firm via the Anthropic and OpenAI APIs, covering 2,400 employees
- Developed a time-series forecasting model to project $1.7M in year-end AI spend, correcting an initial linear model that failed to capture adoption saturation
- Implemented a conversational agent that answers usage/cost questions via natural language by querying a Power BI semantic model hosted on Azure using DAX queries

Machine Learning Researcher — January 2026 – June 2026
Massachusetts Institute of Technology — *Cambridge, MA*

- Benchmarked 4 VLMs across 300+ origami models using IoU and CLIP similarity, surfacing spatial reasoning patterns
- Achieved query efficiency scores of 0.85–0.87 by fine-tuning constraint thresholds on model outputs
- Co-authored OrigamiBench, accepted to ICML 2026 LM4Plan and CVPR 2026 MAR Workshops

Data Science Intern — October 2025 – May 2026
Defense Industrial Base Institute at Northeastern University — *Boston, MA*

- Built an ingestion pipeline to pull DoD contract data from USASpending.gov and geocode vendor locations
- Engineered a Neo4j knowledge graph connecting vendors, contracts, and geographies via UEI/CAGE identifiers
- Transformed 120+ MILCON and OSD PDFs into structured datasets via Python and SQL, cut manual review by 60%.

Graduate Research Assistant — January 2025 – May 2026
Northeastern University — *Boston, MA*

- Engineered Python pipelines for HTML document parsing and CSV generation, reduced prep time by 70%
- Created an AI powered research dashboard using Python and Claude, improved research accessibility by 40%

Data Science Intern — November 2024 – April 2025
Beats Electronics — *Remote*

- Trained a Random Forest model on marketing spend and 1,000 surveys, achieved F1-score of 0.84 for audience targeting
- Developed a TF-IDF NLP sentiment pipeline on 5,000 reviews, identifying customer satisfaction drivers

Projects

Easy Code | Python, Qdrant, Neo4j, AST, RAG | GitHub — June 2026 – Present

- Engineering a hybrid code retrieval system that parses GitHub repositories with AST and indexes code in Qdrant for semantic search
- Building a Neo4j knowledge graph linking files, classes, functions, methods and imports to map codebase structure
- Developing REST API combining vector and graph search to return relevant code with file and line references

Reverse Turing Test | Python, PyTorch, DistilBERT, ONNX, JavaScript | GitHub — February 2026 – May 2026

- Built a 3-model ensemble for AI-generated text detection on 156K samples from the RAID dataset (DistilBERT 98.8%, visual CNN 97.5%, stylometric 76.8%), combined via a stacked meta-classifier for 99.3% accuracy and 0.999 ROC-AUC
- Designed a leakage-free stacking protocol — base models on train, meta-classifier on validation, all metrics on a 12K-sample held-out test split — and benchmarked every model against a 51.5% majority-class baseline
- Quantized both neural models to int8 ONNX (293MB → 71MB) and ported the inference pipeline to JavaScript, running the full ensemble client-side in-browser at 0.4s per classification with no inference server

Technical Skills

Languages: Python, SQL, R, JavaScript
ML/NLP: PyTorch, Hugging Face, scikit-learn, OpenCV, pandas, NumPy, matplotlib, ONNX
LLM/AI: Anthropic API, OpenAI API, RAG, AI Agents, vector search
Tools: Neo4j, Qdrant, Docker, Git, Streamlit, Azure, Power BI, DAX